AI in Finance

Arpit Gupta (NYU Stern)
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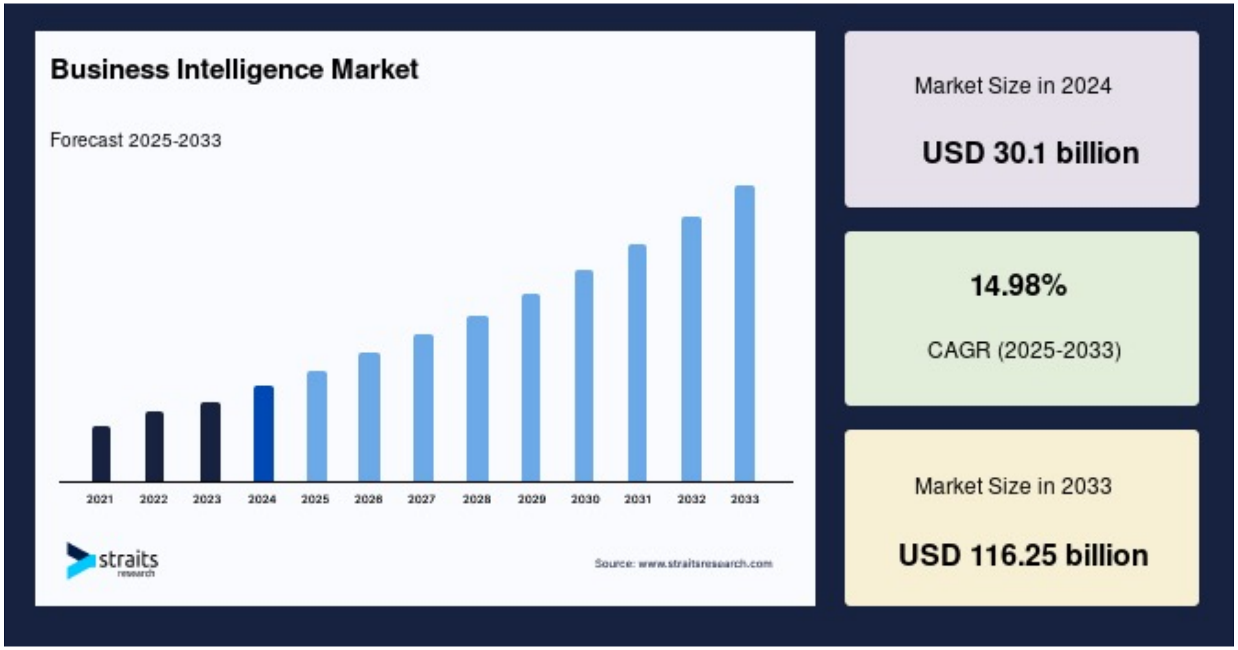
Session 4
MSGF

Impact of Market Intelligence on Finance

- Practice of turning unstructured information (news, filings, earnings calls, regulatory changes, alternative data) into actionable investment decisions
- Provides evidence base for firm strategic decisions, valuation, and market trading



AI Disruption

- The ability for LLMs to process text at scale drastically impacts the core economics of market intelligence
  - This was previously a labor intensive process constrained by human processing ability; now LLMs can lower the bottleneck
- It also enables new categories of simulation and analysis around consumer choices
- But faster and cheaper does not necessarily translate to more valuable:
  - Are speed and processing the binding constraints on investment performance?
  - What is the quality of the information we summarize or simulate?
  - Where does the “alpha” go if everyone has access to the same AI product? Who captures the value if market information is cheaper to process?

Textual Analysis in Finance: Headlines

Headline Analysis and Sentiment

Giving Content to Investor Sentiment: The Role of Media in the Stock Market

Paul C. Tetlock*

ABSTRACT

I quantitatively measure the nature of the media's interactions with the stock market using daily content from a popular *Wall Street Journal* column. I find that high media pessimism predicts downward pressure on market prices followed by a reversion to fundamentals, and unusually high or low pessimism predicts high market trading volume. These results and others are consistent with theoretical models of noise and liquidity traders. However, the evidence is inconsistent with theories of media content as a proxy for new information about fundamental asset values, as a proxy for market volatility, or as a sideshow with no relationship to asset markets.

Table II
Predicting Dow Jones Returns Using Negative Sentiment

The table data come from CRSP, NYSE, and the General Inquirer program. This table shows OLS estimates of the coefficient γ_1 in equation (1). Each coefficient measures the impact of a one standard deviation increase in negative investor sentiment on returns in basis points (one basis point equals a daily return of 0.01%). The regression is based on 3,709 observations from January 1, 1984, to September 17, 1999. I use Newey-West standard errors that are robust to heteroskedasticity and auto-correlation up to five lags. Bold denotes significance at the 5% level; italics and bold denotes significance at the 1% level.

Regressand	Dow Jones Returns		
News Measure	Pessimism	Negative	Weak
$BdNws_{t-1}$	***-8.1***	***-4.4***	***-6.0***
$BdNws_{t-2}$	0.4	**3.6**	2.0
$BdNws_{t-3}$	0.5	-2.4	-1.2
$BdNws_{t-4}$	**4.7**	***4.4***	***6.3***
$BdNws_{t-5}$	1.2	2.9	**3.6**
$\chi^2(5)[Joint]$	***20.0***	***20.8***	***26.5***
p-value	0.001	0.001	0.000
Sum of 2 to 5	**6.8**	***9.5***	***10.7***
$\chi^2(1)[Reversal]$	**4.05**	***8.35***	***10.1***
p-value	0.044	0.004	0.002

For each day in my sample, I gather newspaper data by counting the words in all 77 *predetermined* GI categories from the Harvard psychosocial dictionary. To mitigate measurement error and enhance construct validity, I perform a principal components factor analysis of these categories. This process collapses the 77 categories into a single media factor that captures the maximum variance in the GI categories. Because this single media factor is strongly related to pessimistic words in the newspaper column, I refer to it as a pessimism factor hereafter.

Improved Financial Dictionary

When is a Liability not a Liability?
Textual Analysis, Dictionaries, and 10-Ks

Journal of Finance, forthcoming

TIM LOUGHRAN and BILL MCDONALD

ABSTRACT

Previous research uses negative word counts to measure the tone of a text. We show that word lists developed for other disciplines misclassify common words in financial text. In a large sample of 10-Ks during 1994 to 2008, almost three-fourths of the words identified as negative by the widely used Harvard Dictionary are words typically not considered negative in financial contexts. We develop an alternative negative word list, along with five other word lists, that better reflect tone in financial text. We link the word lists to 10-K filing returns, trading volume, return volatility, fraud, material weakness, and unexpected earnings.

Key words: Textual analysis; Harvard Dictionary; negative word counts; term weighting.

- Existing dictionary approach assigns typical finance words in 10-Ks (“taxes” “liabilities”) as negative, despite context-dependence in Finance (i.e., “liabilities went down”)
- Develop a new finance-specific dictionary

Returns Decline with Usage

Can ChatGPT Forecast Stock Price Movements?

Return Predictability and Large Language Models *

Alejandro Lopez-Lira and Yuehua Tang

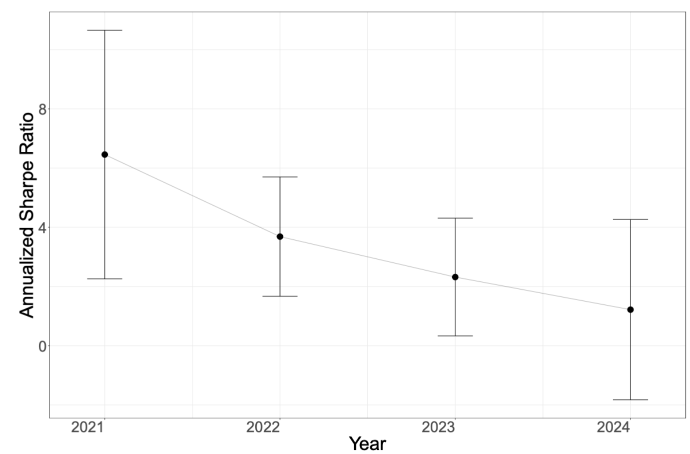
University of Florida

First Version: April 6, 2023; This Version: October 28, 2025

Abstract

We document the capability of large language models (LLMs) like ChatGPT to predict stock market reactions from news headlines without direct financial training. Using post-knowledge-cutoff headlines, GPT-4 captures initial market responses, achieving approximately 90% portfolio-day hit rates for the non-tradable initial reaction. GPT-4 scores also significantly predict the subsequent drift, especially for small stocks and negative news. Forecasting ability generally increases with model size, suggesting that financial reasoning is an emerging capacity of complex LLMs. Strategy returns decline as LLM adoption rises, consistent with improved price efficiency. To rationalize these findings, we develop a theoretical model that incorporates LLM technology, information-processing capacity constraints, underreaction, and limits to arbitrage.

Figure 8: Year-by-Year Performance of Overnight News Strategy: Sharpe Ratios



This figure shows the year-by-year Sharpe ratios of the same long-short strategy based on ChatGPT 4 as in Figure 3. If a piece of news is released before 9 a.m. on a trading day, we enter the position at the market opening and exit at the close of the same day. If the news is announced after the market closes, we assume we enter the position at the next opening price and exit at the close of the next trading day. All the strategies are rebalanced daily. This figure presents the annualized Sharpe ratios of the strategy for four subperiods of our sample (2021Q4, 2022, 2023, and 2024 Jan.-May) and their 95% confidence intervals calculated following Lo (2002).

Generated Beliefs

The Ghost in the Machine: Generating Beliefs with Large Language Models *

J. Leland Bybee

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First Draft: February, 2023
This Draft: February, 2025

Abstract

I introduce a methodology to generate economic expectations by applying large language models to historical news. Leveraging this methodology, I make three key contributions. (1) I show generated expectations closely match existing survey measures and capture many of the same deviations from full-information rational expectations. (2) I use my method to generate 120 years of economic expectations from which I construct a measure of economic sentiment capturing systematic errors in generated expectations. (3) I then employ this measure to investigate behavioral theories of bubbles. Using a sample of industry-level run-ups over the past 100 years, I find that an industry's exposure to economic sentiment is associated with a higher probability of a crash and lower future returns. Additionally, I find a higher degree of feedback between returns and sentiment during run-ups that crash, consistent with return extrapolation as a key mechanism behind bubbles.

Figure 2: Prompt Format

```
Here is a piece of news:

"%s"

Do you think this news will increase or decrease %s?

Write your answer as:

{increase/decrease/uncertain}:
{confidence (0-1)}:
{magnitude of increase/decrease (0-1)}:
{explanation (less than 25 words)}
```

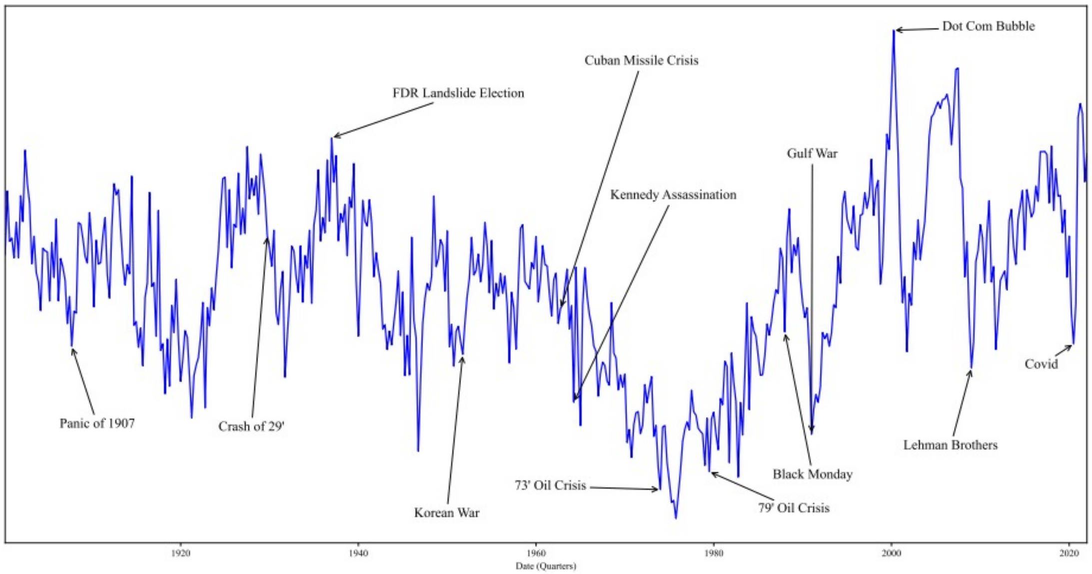
Note. Reports the prompt format for queries made to GPT. "%s" indicates where in the prompt the headline and target text are inserted.

Table 2: Example Responses

Series	Direction	Headline/Response
		(1989-06-29) U.S. Reaches Accords Widening Access To the Mobile Phone Business in Japan
S&P	1	The news signals potential growth opportunities for US mobile phone companies, which could positively impact the S&P 500 index.
UE	-1	Increased access to the Japanese mobile phone market will likely create new job opportunities in the U.S. telecommunications industry.
CPI	-1	Increased competition and access to the Japanese mobile phone market may lead to lower prices for U.S. consumers.
		(2001-02-28) Bush Offers Tax Cuts and Tight Budgets to Aid 'Faltering' Economy
S&P	1	Tax cuts stimulate the economy, which could lead to increased corporate profits and higher stock prices.
UE	0	Tax cuts may stimulate investment, but tight budgets could reduce government spending and slow job growth.
CPI	1	Tax cuts usually stimulate spending, which can increase demand and raise prices, but budget tightening may counteract that effect.

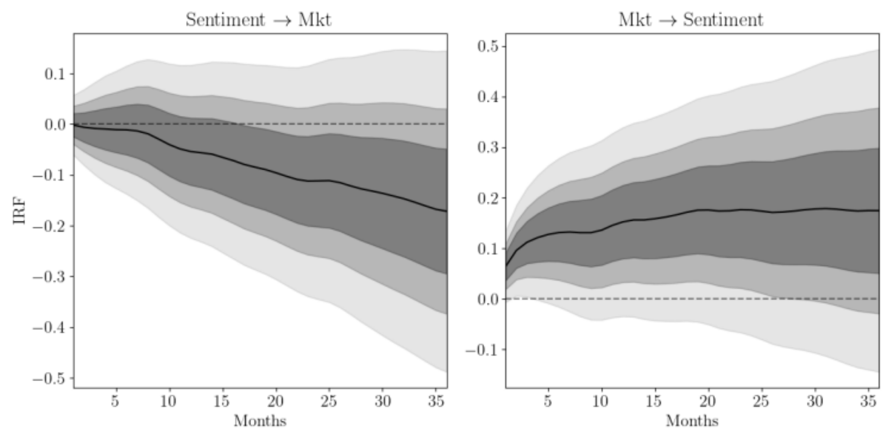
Note. Reports a sample of example article headlines and the corresponding responses from GPT. The first column reports the series queried. The second column reports the direction of the query. The third column reports the headline and the corresponding response from GPT.

Figure 12: Time Series of Economic Sentiment



Note. Reports the quarterly time series of the economic sentiment measure using the first principal component of the ex-ante residuals (EAR) along with labels for key events.

Figure 13: Sentiment and Aggregate Return IRFs



Note. The left panel reports the IRF capturing the response of value-weighted market returns to a one standard deviation shock to the EAR sentiment measure. The right panel reports the IRF capturing the response of the EAR sentiment measure to a one standard deviation shock to value-weighted market returns. 68%, 90%, and 99% confidence intervals are reported (with decreasing levels of shading). Standard errors are Newey-West with the corresponding horizon as the number of lags.

Textual Analysis in Finance: More Complicated

GPT Can Tackle Complex Interpretations

Can ChatGPT Decipher Fedspeak?

Anne Lundgaard Hansen and Sophia Kazinnik*

August 5, 2023

Abstract

Abstract This paper investigates the ability of Generative Pre-training Transformer (GPT) models to decipher Fedspeak, a term used to describe the technical language used by the Federal Reserve to communicate on monetary policy decisions. We evaluate the ability of GPT models to classify the policy stance of Federal Open Market Committee announcements relative to human assessment. We show that GPT models deliver a considerable improvement in classification performance over other commonly used methods. We then demonstrate how the GPT-4 model can provide explanations for its classifications that are on par with human reasoning. Finally, we show that the GPT-4 model can be used to identify macroeconomic shocks using the narrative approach of Romer and Romer (1989, 2023).

Table 5: Explanations of classifications of **Sentence 1** as provided by the human research assistant (Bryson) and the GPT-3 and GPT-4 models.

Sentence 1: *In light of the current shortfall of inflation from 2 percent, the committee will carefully monitor actual and expected progress toward its inflation goal.*

Name	Label	Explanation
Bryson	Dovish	This sentence emphasizes the current shortfall of inflation below the Committee's target, suggesting that loose monetary (low FFR or securities purchases) policies will be necessary to bring inflation up towards the Committee's target.
GPT-3	Neutral	This sentence states that the committee will monitor progress towards its inflation goal, without leaning towards any particular policy stance.
GPT-4	Mostly dovish	The sentence emphasizes the shortfall of inflation from the target and the committee's intention to monitor progress, suggesting a cautious approach and potential inclination towards easing monetary policy.

Representations of Firms

Economic Representations

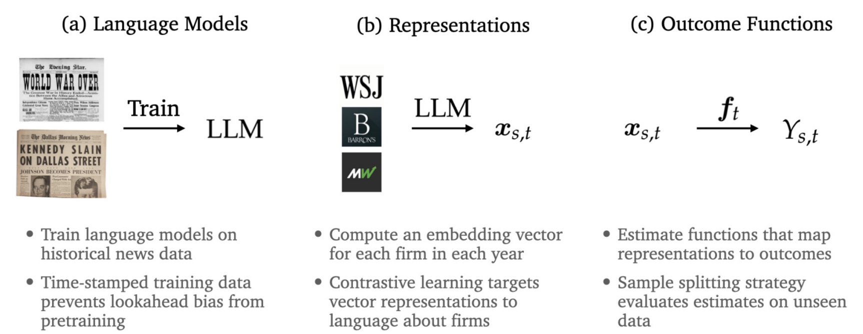
Suproteem K. Sarkar

January 11, 2025

Please click here for the latest version

Valuations depend on how people categorize, perceive, or otherwise *represent* economic objects. This paper develops a measure of how the market represents firms, and uses this measure to study stock valuations. I train an algorithm to structure language from financial news into embeddings—vectors that quantify the economic features and themes in each firm's news coverage. I show that a firm's vector representation is informative of how the market perceives its business model. Representations explain cross-sectional variation in stock valuations, cash flow forecasts, and return correlations. Changes in representation help to explain changes in stock prices. Some changes in representations and prices are forecastable, and indicate that some of the explained variation in stock valuations stems from misperception. I find that misperception and misvaluation can intensify when a firm's news coverage includes attention-drawing features—like "internet" in the late 1990s or "AI" in the early 2020s.

Figure 1: Summary of empirical procedures.



Notes: This figure summarizes the empirical procedures in this paper. First, I train language models on historical newspaper data—this avoids lookahead bias, which can be an issue with off-the-shelf language models. Second, I develop an algorithm to transform financial news language into vector representations for each firm in each year. Third, I estimate functions that relate representations to a series of outcome variables using a split-sample estimation strategy.

Figure 2: Representations encode relations between firms.

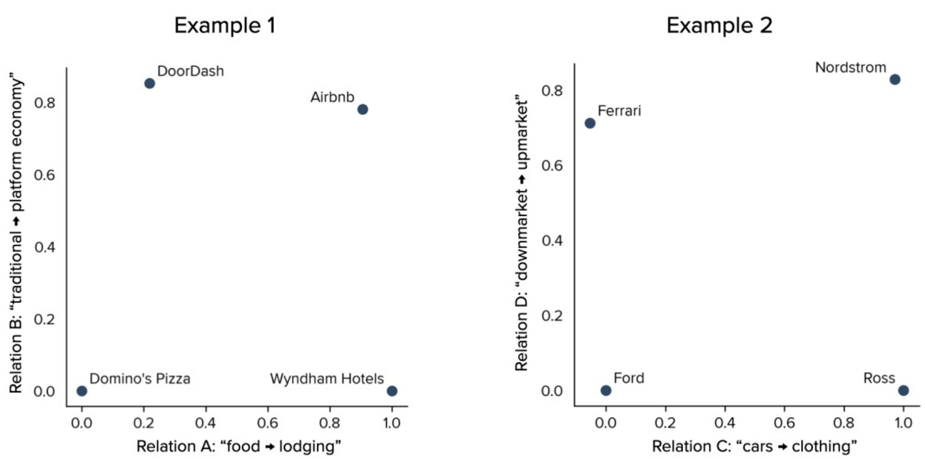
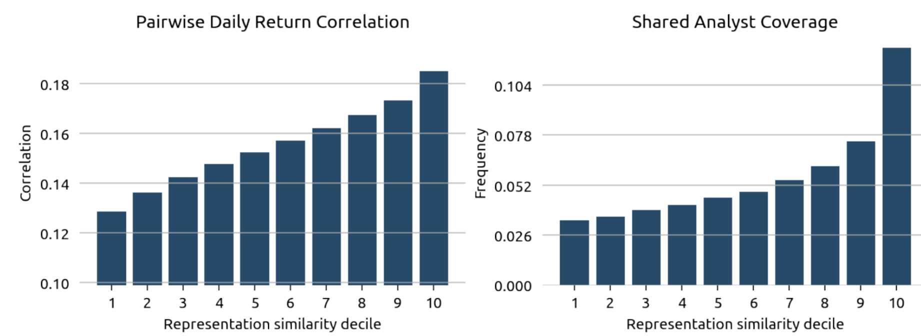


Figure 3: Representations encode similarity between firms.



Notes: This figure plots the relationship between representation similarity and standard measures of market similarity. The horizontal axis in each subplot corresponds to the decile of pairwise representation similarity between firms. The vertical axis corresponds to the pairwise daily return correlation or frequency of firms with shared analyst coverage.

Table 1: Representations help to explain prices and forecasts.

R^2: Estimates of price and forecast variables			
Dependent variable →	log(P/B)	log(P/E)	$\mathbb{F}$[LTG]
Explanatory variables ↓	(1)	(2)	(3)
Representation	26.0%	12.1%	19.7%
Industry (FF12)	9.2%	6.8%	12.8%
Sub-Industry (FF48)	11.7%	7.5%	13.6%
Characteristics	12.8%	6.9%	22.2%
Representation+All Others	32.9%	14.5%	30.1%
All Others	19.3%	10.7%	25.7%

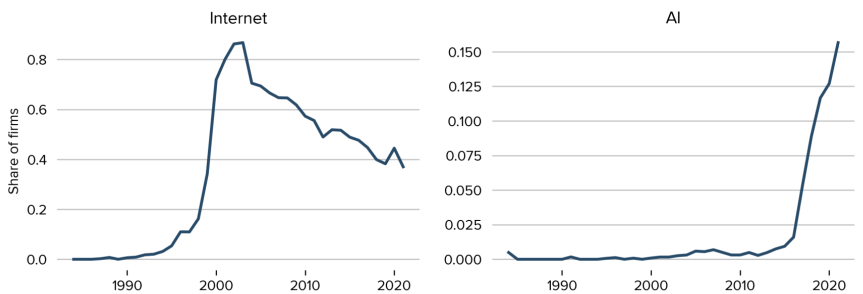
Notes: This table reports the R^2 on a series of estimates of stock price and cash-flow forecast variables across sets of explanatory variables. The estimates are from Equation (5): $Y_{s,t} = \widehat{\boldsymbol{v}}_t \cdot \boldsymbol{x}_{s,t} + \varepsilon_{s,t}$, following the split-sample approach of Section 2.3. Each row reports R^2 estimates across a different set of explanatory variables—representations, industry codes, and characteristics, and concatenations of these vectors. Each column reports R^2 estimates across a different price or forecast variable. The set of firms used to fit each estimator does not include the firms used to evaluate the estimates.

Table 2: Representations help to explain returns.

R^2: Estimates of annual returns			
Estimation strategy →	[Δv]	[Δx]	[Total]
Explanatory variables ↓	(1)	(2)	(3)
Representation	9.4%	6.8%	13.4%
Industry (FF12)	3.5%	$<$0%	3.5%
Sub-Industry (FF48)	4.1%	$<$0%	4.1%
Characteristics	3.1%	4.4%	8.6%
Representation+All Others	11.5%	10.0%	18.7%
All Others	6.4%	4.4%	11.3%

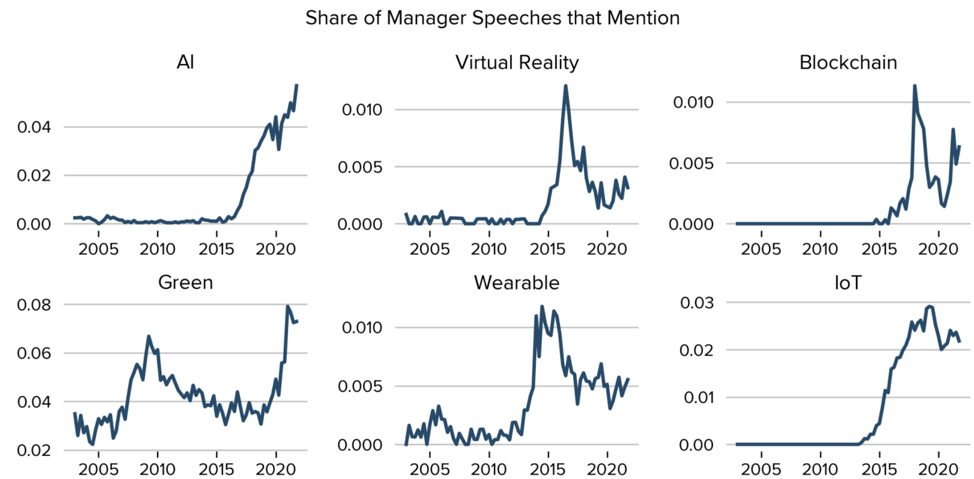
Notes: This table reports split-sample R^2 statistics of using estimated returns to explain realized cross-sectional returns, following Equation (6): $R_{s,t+1} = \widehat{\mathbb{E}}_t[R_{s,t+1} \mid z] + \varepsilon_{s,t+1}$. Each row reflects a different set of explanatory variables: Representations, industry vectors, characteristics vectors, and combinations of these vectors. Each column reflects a different estimation strategy.

Figure 4: Motivating evidence of attention-drawing features: Firm-level word mentions.



Notes: This figure plots the share of firms whose news coverage includes a given word in a given year. The two panels report these statistics for the words "internet" and "AI." Internet began to have a large increase in attention in the late 1990s, and AI began to have a large increase in attention in the late 2010s. During these periods of increases in attention, case study evidence suggests that the market neglected fundamental features of these firms (Cooper et al., 2001; Narayanan and Kapoor, 2024). Figure A2 also includes statistics for "virtual reality," "blockchain," "green," and "wearable."

Figure 5: Communication across attention-drawing features.



Notes: This figure plots the share of earnings call speeches by managers that include a given phrase in a given quarter. Each of these phrases had a sharp increase in managerial mentions at least once over the sample. For some of these phrases, sharp increases in managerial mentions were followed by sharp decreases in managerial mentions.

AI Escalating Complexity in Textual Analysis

- Each step of progression shows improved ability to handle nuanced economic and financial language
  - From sentiment analysis on headlines with keywords, to interpretations of headlines, to analysis of paragraphs and statements, to sophisticated representations of entire firms and businesses
- Each step unlocks a different layer of economic value, and unleashes a different set of competitive responses
  - The keyword level is cheap, reproducible, and widely available.
    Easily arb-ed away
  - Headline interpretation also easy (and we see alpha decaying)
  - Document and firm analysis approaches what a senior analyst does

Alternative Data

Big Data from Space

On the Capital Market Consequences of Big Data: Evidence from Outer Space

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Abstract

We use the introduction of satellite coverage of major retailers to study the capital marke implications of unequal access to big data. Satellite data enabled sophisticated investors wit access to such data to formulate profitable trading strategies, especially by targeting th upcoming reports of retailers with bad news for the quarter. The introduction of satellit data led to more informed short-selling activity, less informed individual buying activity, an lower stock liquidity around the reports of retailers with satellite coverage. We conclude tha unequal access to big data can increase information asymmetry among market participant without immediately enhancing price discovery.

FIGURE 2

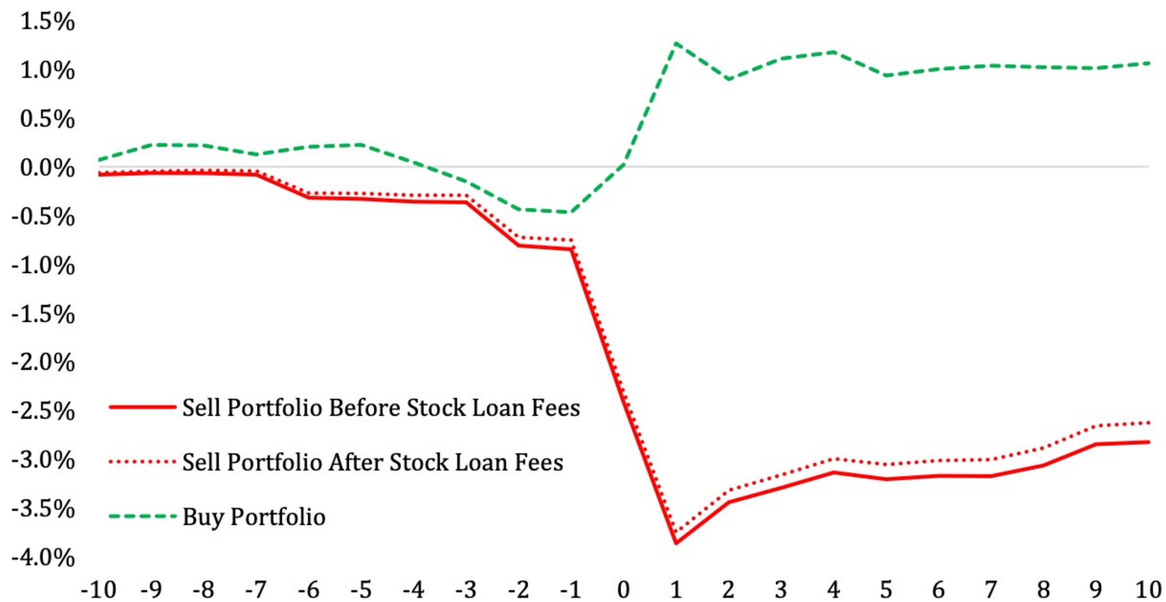
Illustrative Example of Satellite Imagery

This figure illustrates the measurement of key variables using satellite imagery data for Target Corporation, the department store company. The satellite image is for the Target store located at 4500 Macdonald Ave, Richmond CA 94805. The image was captured on September 19, 2016, at 11:03am. The processed image indicates the number of cars present within a fixed area of parking lot spaces that RS Metrics assigns to each store. The yellow line outlines the boundary of the parking lot associated with Target and the red dots indicate the occupied parking lot spaces. The parking lot spaces assigned to each store do not change over time unless the company renovates the parking lot. At the time of the satellite image, RS Metrics reports 540 parking lot spaces with 146 of them filled. The parking lot spaces on the bottom right of this Target store are excluded because they may represent employee parking. As a general rule for any individual store location, RS Metrics defines the "most likely parking area" for customers and keeps that parking lot boundary relatively fixed over time so that the variability in the data comes from the number of cars parked at any time. Starting with the granular parking lot data for Target Corporation in 2016:Q3, we identify 1,210 individual store locations across the U.S. with year-over-year satellite coverage, i.e., coverage in both 2016:Q3 and 2015:Q3. We calculate the average parking lot size and parking lot traffic per Target store during the quarter, and we sum across stores to obtain the enterprise-level information. For 2016:Q3 across the 1,210 Target store locations with year-over-year satellite coverage, the aggregate parking lot traffic is 156,977 while the aggregate parking lot space is 595,340. It follows that the parking lot fill rate for Target Corporation in 2016:Q3 is 26.37%. Repeating the steps for 2015:Q3, we find a fill rate of 26.94%. Hence, the year-over-year growth rate in the fill rate is −2.14%.

FIGURE 3
Formulating a Trading Strategy Using Satellite Imagery Data

This figure presents the cumulative factor-adjusted return from a strategy that buys (short sells) retailers with same-store parking lot fill rate growth in the top (bottom) quartile of the cross-sectional distribution of $\Delta FLRT_{iq}$. The measurement window is from ten trading days before to ten trading days after the quarterly earnings announcement (day zero). In Panel A, the green dashed (red solid) line presents the performance of the portfolio that buys (short sells) retailers in the top (bottom) quartile portfolio of $\Delta FLRT_{iq}$, while the red dotted line presents the performance of the short-sell portfolio net of stock loan fees. In Panel B, we present the daily abnormal share turnover for the buy and short-sell portfolios. We measure share turnover as the daily trading volume divided by the number of shares outstanding. To obtain the abnormal values of share turnover, we adjust the daily values of share turnover during the event window for the daily share turnover prior to the event window. The sample includes 650 firm-quarter observations from 2011:Q1 to 2017:Q4.

Panel A: Buy and Sell Portfolio Returns.



Shifts Attention to Shorter Horizons

Does Alternative Data Improve Financial Forecasting?

The Horizon Effect

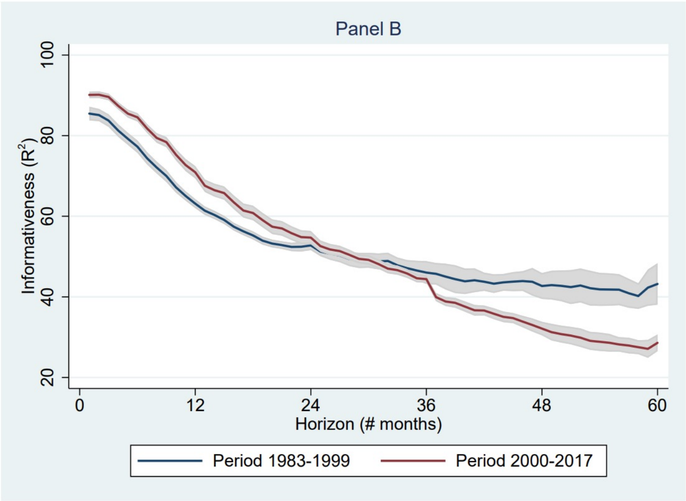
Journal of Finance, forthcoming

OLIVIER DESSAINT, THIERRY FOUCAULT,

and LAURENT FRESARD*

ABSTRACT

Existing research suggests that alternative data is mainly informative about short-term future outcomes. We show theoretically that the availability of short-term oriented data can induce forecasters to optimally shift their attention from the long-term to the short-term because it reduces the cost of obtaining short-term information. Consequently, the informativeness of their long-term forecasts decreases, even though the informativeness of their short-term forecasts increases. We test and confirm this prediction by considering how the informativeness of equity analysts' forecasts at various horizons varies over the long run and with their exposure to social media data.



This figure displays the term structure of analysts' forecast informativeness. Each graph shows the means of analyst-level $R^2_{i,t,h}$ over all analysts and dates, for fixed h values expressed in number of months (displayed on the x-axis). The forecasting horizon h is measured as the number of days between the forecasting date and the date of actual earnings release, divided by 365. The sample period is 1983-2017 (Panel A), split into two sub-periods (Panel B). The shaded gray area corresponds to a 90% confidence interval, obtained with standard errors clustered by forecasting fiscal period.

Shifts Focus to Text Analyzable Data

Artificial Intelligence Tools Expand Scientists' Impact but Contract Science's Focus

(Just accepted by *Nature*, to be online soon)

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ABSTRACT

Development in Artificial Intelligence (AI) has accelerated scientific discovery[1]. Alongside recent AI-oriented Nobel prizes[2–9], these trends establish the role of AI tools in science[10]. This advancement raises questions about the potential influences of AI tools on scientists and science as a whole, and highlights a potential conflict between individual and collective benefits[11]. To evaluate, we used a pretrained language model to identify AI-augmented research, with an F1-score of 0.875 in validation against expert-labeled data. Using a dataset of 41.3 million research papers across natural science and covering distinct eras of AI, here we show an accelerated adoption of AI tools among scientists and consistent professional advantages associated with AI usage, but a collective narrowing of scientific focus. Scientists who engage in AI-augmented research publish 3.02 times more papers, receive 4.84 times more citations, and become research project leaders 1.37 years earlier than those who do not. By contrast, AI adoption shrinks the collective volume of scientific topics studied by 4.63% and decreases scientist's engagement with one another by 22.00%. Thereby, AI adoption in science presents a seeming paradox—an expansion of individual scientists' impact but a contraction in collective science's reach—as AI-augmented work moves collectively toward areas richest in data. With reduced follow-on engagement, AI tools appear to automate established fields rather than explore new ones, highlighting a tension between personal advancement and collective scientific progress.

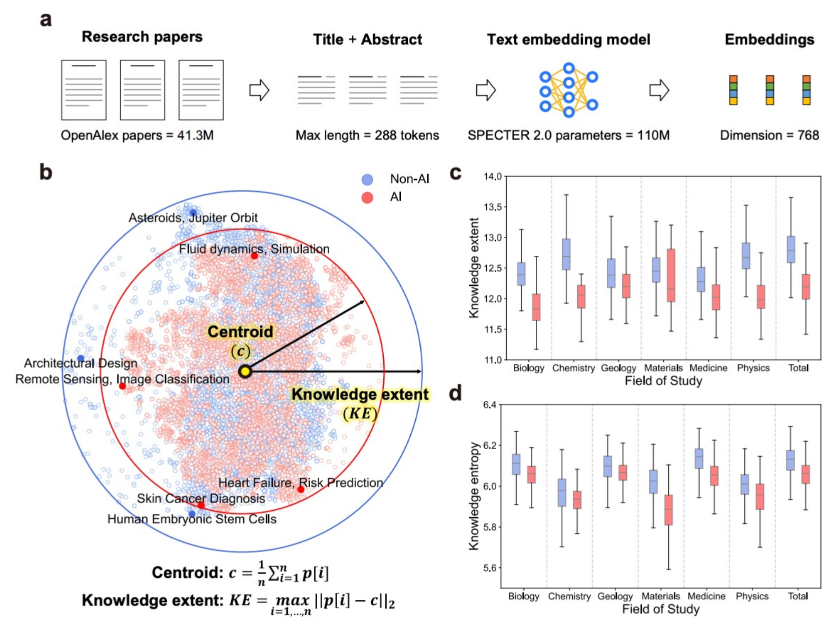


Figure 3. AI adoption is associated with a contraction in knowledge extent within and across scientific fields. (a) We embed research papers into a 768-dimensional vector space with a pre-trained text embedding model, then measure the knowledge extent of papers within that space. **(b)** For visualization, we use the t-SNE algorithm to flatten the high-dimensional embeddings of a random batch of 10,000 papers, half of which are AI papers and half are non-AI papers, into a 2-D plot. As shown by the solid arrows and circular boundaries, the knowledge extent of AI papers (calculated in the unflatted space) is smaller across the entirety of natural science, and AI papers are more clustered in knowledge space, indicating more concentration on specific problems. **(c)** Knowledge extent of AI and non-AI papers in each field ($p < 0.001, n = 1,000$ samples in each field), where AI research focuses on a more contracted knowledge space. **(d)** Knowledge entropy of AI and non-AI papers in each field ($p < 0.001, n = 1,000$ samples in each field), where AI research has a lower entropy. For panels (c) and (d), boxplots are centred at the median and bounded at the first and third quartile (Q1 and Q3), with 1.5 times of the inter-quartile range (IQR) shown as whiskers from the box. All statistical tests use a median-test.

Man vs Machine

A B S T R A C T

An AI analyst trained to digest corporate disclosures, industry trends, and macroeconomic indicators surpasses most analysts in stock return predictions. Nevertheless, humans win "Man vs. Machine" when institutional knowledge is crucial, e.g., involving intangible assets and financial distress. AI wins when information is transparent but voluminous. Humans provide significant incremental value in "Man + Machine", which also substantially reduces extreme errors. Analysts catch up with machines after "alternative data" become available if their employers build AI capabilities. Documented synergies between humans and machines inform how humans can leverage their advantage for better adaptation to the growing AI prowess.

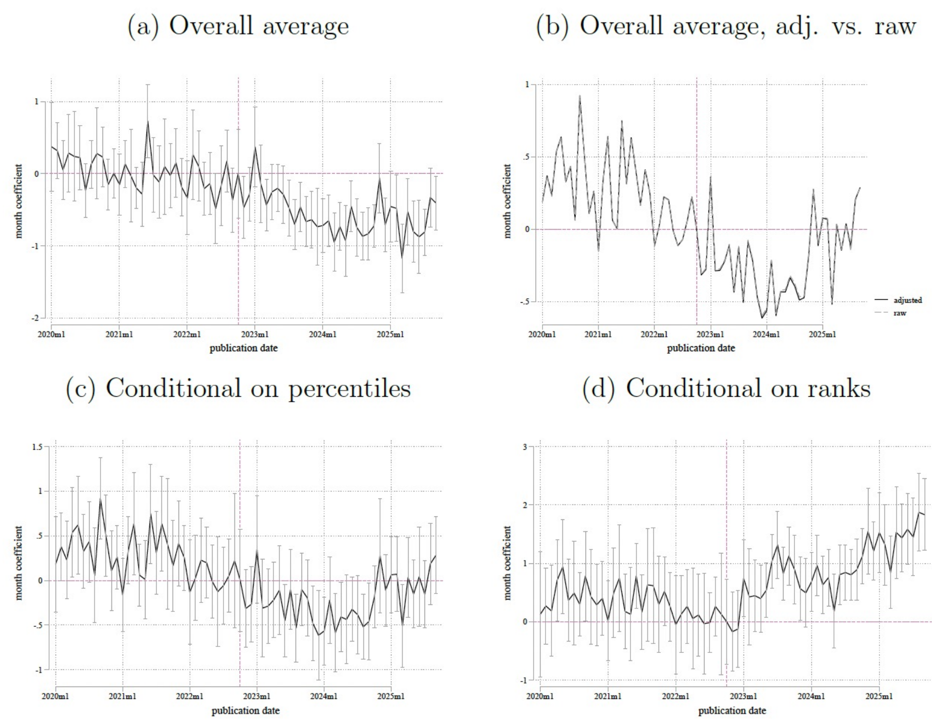
More Valuable Books?

AI and the Quantity and Quality of Creative Products: Have LLMs Boosted Creation of Valuable Books?
Imke Reimers and Joel Waldfogel
NBER Working Paper No. 34777
January 2026
JEL No. L16, L82

ABSTRACT

With the diffusion of LLMs between 2022 and 2025, new book releases have tripled, raising a question of AI's impact on book quality. We develop a ratings-based usage measure that is comparable across book release vintages, and we find that the vintages from the AI influx period have lower average quality. Yet, the top 1,000 monthly releases per category - albeit not the top 100 - have higher quality than before; and the effect is larger in categories with faster growth in new titles. Authors entering since the LLM influx produce predominantly low-quality work; and the higher-quality output of pre-LLM authors entrants has risen. A nested logit calibration shows that LLM-enhanced book production could, in steady state, raise the surplus that consumers derive from book markets by a quarter to a half.

Figure 6: Quality effects: "high-low" approach



Notes: The figures report the coefficients on monthly adjusted usage effects for high-entry-growth categories relative to the others. Panel (a) reports the average effect. Panel (b) compares effects using the adjusted vs unadjusted usage measure. Panel (c) reports effects conditional on rank percentiles, and Panel (d) reports effects conditional on rank positions (1-100, etc). All regressions include category fixed effects and month effects, and standard errors are clustered on category × release month.

Economic Implications

- Commodification of the reading layer
  - Everyone can now process the same documents at the same speed, so the alpha shifts to who can interpret it best and act on it most effectively
  - i.e., impact of Bloomberg on democratizing access to market data
- Attention Reallocation Problem
  - At least for the time being, possible role for analysts to focus on longer-term horizon impacts that are harder for AI to forecast
- Adversarial Response
  - The subject of analysis: corporate management teams, central bankers, will adapt to knowing their words are being machine-parsed
    - i.e., Cook et al. (2023) find banks strategically adjusted the language of their earnings calls in 2023 with stressed/healthy banks converging on similar text

The Astronomer (Vermeer)

Simulations

Homo Silicus

Large Language Models as Simulated Economic Agents: What Can We Learn from Homo Silicus?
John J. Horton
NBER Working Paper No. 31122
April 2023
JEL No. D0

ABSTRACT

Newly-developed large language models (LLM)—because of how they are trained and designed—are implicit computational models of humans—a homo silicus. LLMs can be used like economists use homo economicus: they can be given endowments, information, preferences, and so on, and then their behavior can be explored in scenarios via simulation. Experiments using this approach, derived from Charness and Rabin (2002), Kahneman, Knetsch and Thaler (1986), and Samuelson and Zeckhauser (1988) show qualitatively similar results to the original, but it is also easy to try variations for fresh insights. LLMs could allow researchers to pilot studies via simulation first, searching for novel social science insights to test in the real world.

General Social Agents*

Benjamin S. Manning, MIT
John J. Horton, MIT & NBER

September 24, 2025

MOST RECENT DRAFT [HERE]

Abstract

Useful social science theories predict behavior across settings. However, applying a theory to make predictions in new settings is challenging: rarely can it be done without ad hoc modifications to account for setting-specific factors. We argue that AI agents put in simulations of those novel settings offer an alternative for applying theory, requiring minimal or no modifications. We present an approach for building such "general" agents that use theory-grounded natural language instructions, existing empirical data, and knowledge acquired by the underlying AI during training. To demonstrate the approach in settings where no data from that data-generating process exists—as is often the case in applied prediction problems—we design a heterogeneous population of 883,320 novel games. AI agents are constructed using human data from a small set of conceptually related but structurally distinct "seed" games. In preregistered experiments, on average, agents predict initial human play in a random sample of 1,500 games from the population better than (i) a cognitive hierarchy model, (ii) game-theoretic equilibria, and (iii) out-of-the-box agents. For a small set of separate novel games, these simulations predict responses from a new sample of human subjects *better* even than the most plausibly relevant published human data.

Generative Agent Simulations

Generative Agent Simulations of 1,000 People

Authors: Joon Sung Park[1]*, Carolyn Q. Zou[1,2], Aaron Shaw[2], Benjamin Mako Hill[3], Carrie Cai[4], Meredith Ringel Morris[5], Robb Willer[6], Percy Liang[1], Michael S. Bernstein[1]

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Abstract:

The promise of human behavioral simulation—general-purpose computational agents that replicate human behavior across domains—could enable broad applications in policymaking and social science. We present a novel agent architecture that simulates the attitudes and behaviors of 1,052 real individuals—applying large language models to qualitative interviews about their lives, then measuring how well these agents replicate the attitudes and behaviors of the individuals that they represent. The generative agents replicate participants' responses on the General Social Survey 85% as accurately as participants replicate their own answers two weeks later, and perform comparably in predicting personality traits and outcomes in experimental replications. Our architecture reduces accuracy biases across racial and ideological groups compared to agents given demographic descriptions. This work provides a foundation for new tools that can help investigate individual and collective behavior.

Digital Twins

Database Report: Twin-2K-500: A Data Set for Building Digital Twins of over 2,000 People Based on Their Answers to over 500 Questions

Olivier Toubia, George Z. Gui, Tianyi Peng, Daniel J. Merlau, Ang Li, Haozhe Chen

Published Online: 20 Aug 2025 | https://doi.org/10.1287/mksc.2025.0262

Abstract

Large language model (LLM)-based digital twin simulation, where LLMs are used to emulate individual human behavior, holds great promise for research in business, artificial intelligence, social science, and digital experimentation. However, progress in this area has been hindered by the scarcity of real individual-level data sets that are both large and publicly available. To address this gap, we introduce a large-scale public data set designed to capture a rich and holistic view of individual human behavior. We survey a representative sample of $N = 2,058$ participants (average 2.42 hours per person) in the United States across four waves with more than 500 questions in total, covering a comprehensive battery of demographic, psychological, economic, personality, and cognitive measures, as well as replications of behavioral economics experiments and a pricing survey. The final wave repeats tasks from earlier waves to establish a test-retest accuracy baseline. Initial analyses suggest the data are of high quality and show promise for constructing digital twins that predict human behavior well at the individual and aggregate levels. Beyond LLM applications, due to its unique breadth and scale, the data set also enables broad social science and business research, including studies of cross-construct correlations and heterogeneous treatment effects.

A MEGA-STUDY OF DIGITAL TWINS REVEALS STRENGTHS, WEAKNESSES AND OPPORTUNITIES FOR FURTHER IMPROVEMENT

ABSTRACT

Do ‘digital twins’ capture individual responses in surveys and experiments? We run 19 pre-registered studies on a national U.S. panel and their digital twins (constructed based on

previously-collected extensive individual-level data) and compare twin and human answers across 164 outcomes. The correlation between twin and human answers is modest (approximately 0.2 on average) and twin responses are less variable than human responses. While constructing digital twins based on rich individual-level data improves our ability to capture heterogeneity across participants and predict relative differences between them, it does not substantially improve our ability to predict the exact answers given by specific participants or enhance predictions of population means. Twin performance varies by domain and is higher among more educated, higher-income, and ideologically moderate participants. These results suggest current digital twins can capture some degree of relative differences but are unreliable for individual-level predictions and sample mean and variance estimation, underscoring the need for careful validation before use. Our data and code are publicly available for researchers and practitioners interested in optimizing digital twin pipelines.

Silicon Samples

Out of One, Many:
Using Language Models to Simulate Human Samples

Lisa P. Argyle[1], Ethan C. Busby[1], Nancy Fulda[2], Joshua Gubler[1], Christopher Rytting[2], and David Wingate[2]

[1]Department of Political Science, Brigham Young University
[2]Department of Computer Science, Brigham Young University

September 16, 2022

Abstract

We propose and explore the possibility that language models can be studied as effective proxies for specific human sub-populations in social science research. Practical and research applications of artificial intelligence tools have sometimes been limited by problematic biases (such as racism or sexism), which are often treated as uniform properties of the models. We show that the "algorithmic bias" within one such tool– the GPT-3 language model– is instead both fine-grained and demographically correlated, meaning that proper conditioning will cause it to accurately emulate response distributions from a wide variety of human subgroups. We term this property *algorithmic fidelity* and explore its extent in GPT-3. We create "silicon samples" by conditioning the model on thousands of socio-demographic backstories from real human participants in multiple large surveys conducted in the United States. We then compare the silicon and human samples to demonstrate that the information contained in GPT-3 goes far beyond surface similarity. It is nuanced, multifaceted, and reflects the complex interplay between ideas, attitudes, and socio-cultural context that characterize human attitudes. We suggest that language models with sufficient algorithmic fidelity thus constitute a novel and powerful tool to advance understanding of humans and society across a variety of disciplines.

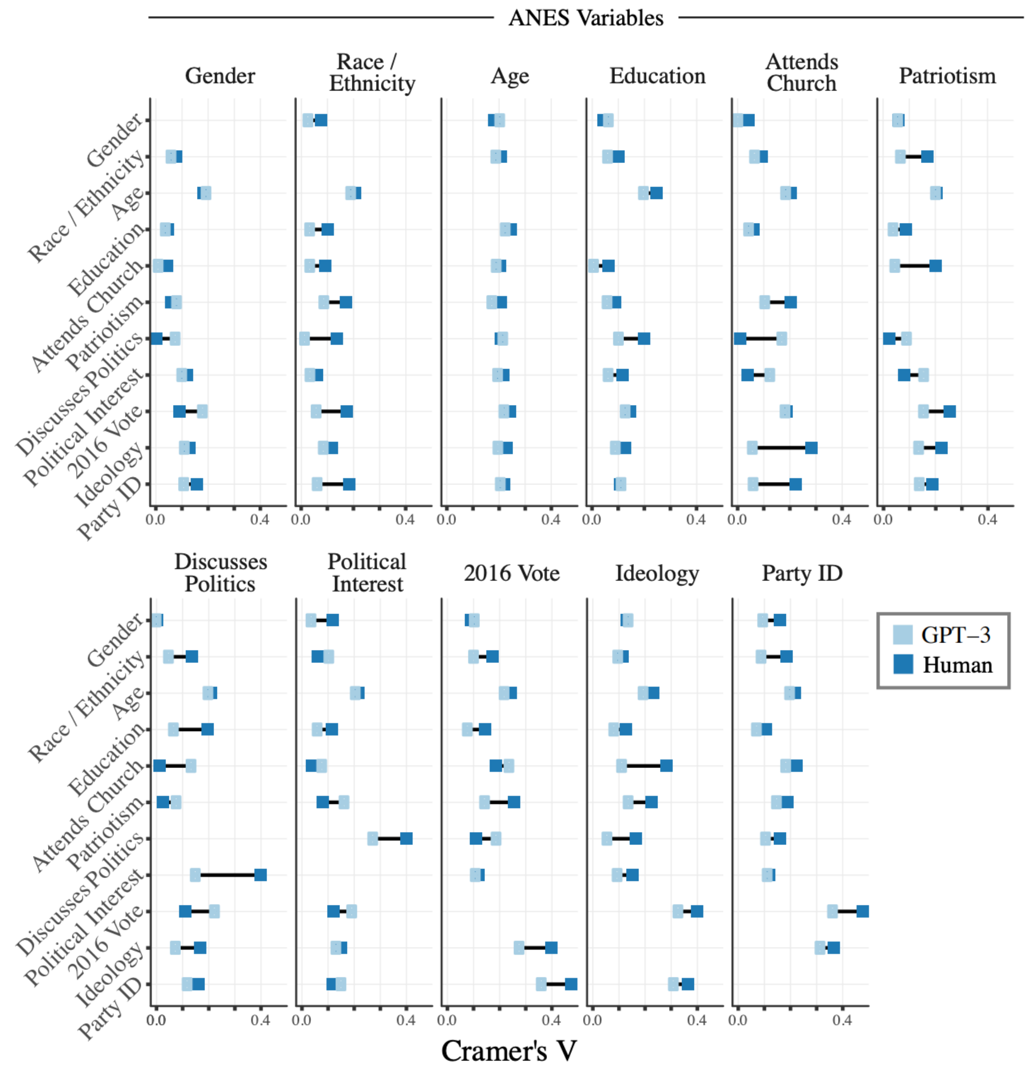


Figure 4: Cramer's V Correlations in ANES vs. GPT-3 Data

Sampled completions

Hillary Clinton. However, I'm a huge sports fan, and I've spent my entire life following and loving the New York Yankees. I'm also a huge fan of the New York Giants. I...

Bernie Sanders in the primaries. I am a member of the Democratic Socialists of America. But as a Jewish American, I will not be voting for the Democratic...

Hillary Clinton. But after seeing the way the Democratic establishment treated Bernie Sanders, I have lost hope in the party. In 2018, I joined the Democratic Socialists of...

Hillary Clinton, because I believed in her vision for America. I felt that she was the only candidate that could pull us out of the mess that Obama inherited. I am a loyal...

Bernie Sanders. I am pro-LGBTQ. But I am also pro-life. I am Jewish. I have never been religious. I was raised in a secular household, but my Jewishness is intrinsic to me. I...

Context

I am a strong conservative. I support traditional values. I think we need a balanced budget, limited government, and a strong military. I'm opposed to abortion and I'm skeptical of global warming. In 2016, I voted for

I am a strong liberal. I support progressive values. I support open immigration and women's right to choose. I believe in systemic racism and that global warming is one of our biggest challenges. In 2016, I voted for

Sampled completions

Donald Trump, but I didn't vote for him because I agreed with him on all the issues. I voted for him because I didn't want Hillary Clinton to be president. I disagreed with many...

John Kasich. I haven't changed my mind on any of those issues. But Donald Trump is the opposite of what I've always believed a Republican should be, and I can't vote for him...

Donald Trump. In other words, I'm a typical Republican voter, except that I'm gay. And that's why I'm leaving the Republican Party. I've felt increasingly unwelcome in the...

Donald Trump. And I have no regrets. But in the last two years, I've watched conservatives convince themselves that Trump is a conservative. And this is a lie. Trump has...

Donald Trump. I consider myself a Republican. But what does that mean these days? I'm not so sure. Which brings me to this poll, released today by CBS News and the New York...

Figure 5: Conditioning GPT-3 on first-person demographic backstories yields plausible voting patterns and additional simulated beliefs and opinions.

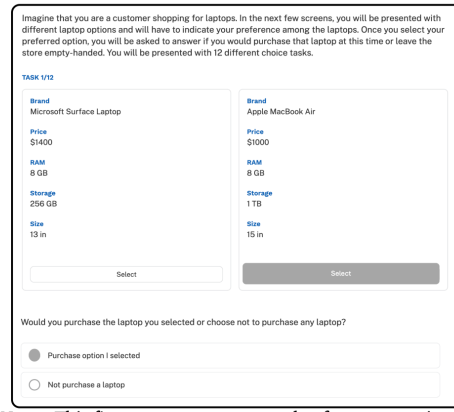
Fine-Tuning

Using LLMs for Market Research

Abstract

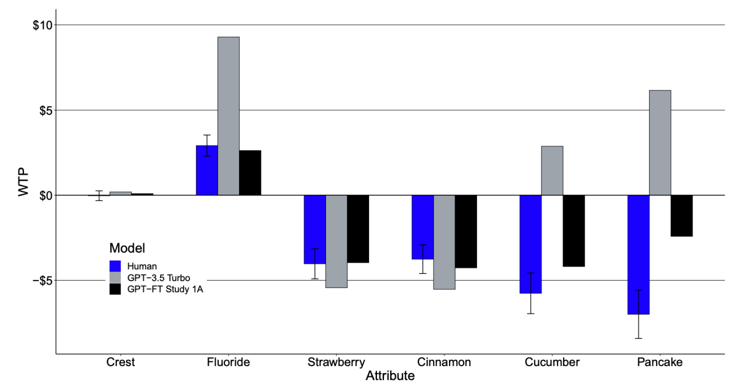
Large language models (LLMs) have rapidly gained popularity as labor-augmenting tools for programming, writing, and many other processes that benefit from quick text generation. In this paper we explore the uses and benefits of LLMs for researchers and practitioners who aim to understand consumer preferences. We focus on the distributional nature of LLM responses, and query the different LLMs to generate dozens of responses to each survey question. We offer two sets of results to illustrate and assess our approach. First, we show that estimates of willingness-to-pay for products and features derived from LLM responses are realistic and comparable to estimates from human studies. Second, we demonstrate a practical method for market researchers to enhance LLMs' responses by incorporating previous survey data from similar contexts via fine-tuning. This method improves the alignment of the LLM responses with human responses for existing and, importantly, new product features. We do not find similar improvements in the alignment for new product categories or for differences between customer segments.

Figure 1: Converting Human Survey Response to Fine-tuning Input Material



{"**messages**": [
{"**role**": "system",
"**content**": "You are a customer..." },
{"**role**": "user",
"**content**": "While Shopping, you see...
1. Microsoft Surface Laptop, Price:$1400,
RAM:8GB, Storage:256GB, Size:13in
2. Apple MacBook Air, Price:$1000,
RAM:8GB, Storage:1000GB, Size:15in...
Did you purchase either of the available
laptops? If so, which one?"},
{"**role**": "assistant",
"**content**": "15in MacBook for $1000,
1000GB of storage, 8GB of RAM"}
]}

Notes: This figure presents an example of our conversion from the human study responses to data which can be provided to GPT for fine-tuning. The human respondent in this case chose to purchase the Macbook Air, which we convert to a text response in the assistant's "content" field to send to GPT. Our fine-tuning data, on the right hand side, includes our description of the setting (a customer who is randomly selected while shopping), the available products, and the customer's choice (complete system and user prompts appear in Web Appendix E).



(b) Study 1B: Toothpaste with New Flavors

Notes: Estimated WTP for humans and LLMs for Studies 1A and 1B. Brackets indicate 95% confidence intervals on our estimates from the human study. GPT-FT represents a fine-tuned GPT model.

Simulating Demand

The Challenge of Using LLMs to Simulate Human Behavior:
A Causal Inference Perspective

George Gui and Olivier Toubia*

November 22, 2025

Abstract

Large Language Models (LLMs) have shown impressive potential to simulate human behavior. We identify a fundamental challenge in using them to simulate experiments: when LLM-simulated subjects are blind to the experimental design (as is standard practice with human subjects), variations in treatment systematically affect unspecified variables that should remain constant, violating the unconfoundedness assumption. Using demand estimation as a context and an actual experiment with 40 different products as a benchmark, we show this can lead to implausible results. While confounding may in principle be addressed by controlling for covariates, this can compromise ecological validity in the context of LLM simulations: controlled covariates become artificially salient in the simulated decision process. We show formally that confoundness stems from ambiguous prompting strategies. Therefore, it can be addressed by developing unambiguous prompting strategies through unblinding, i.e., revealing the experiment design in LLM simulations. Our empirical results show that this strategy consistently enhances model performance across all tested models, including both out-of-box reasoning and non-reasoning models. We also show that it is a technique that complements fine-tuning: while fine-tuning can improve simulation performance, an unambiguous prompting strategy makes the predictions robust to the inclusion of irrelevant data in the fine-tuning process.

Prompt 2: Ask Purchase

System: You, AI, are a customer. Your task is to fill in the blanks ______. Return the completed information in comma-separated values, without any extra text.

User: Please consider the following product category: {category}.

Suppose you are in a grocery store, and you see the following product in that category: {product}.

The product is currently priced at ${Price}. Would you or would you not purchase the product? ______ ["purchase" or "not purchase"]

Return example: purchase

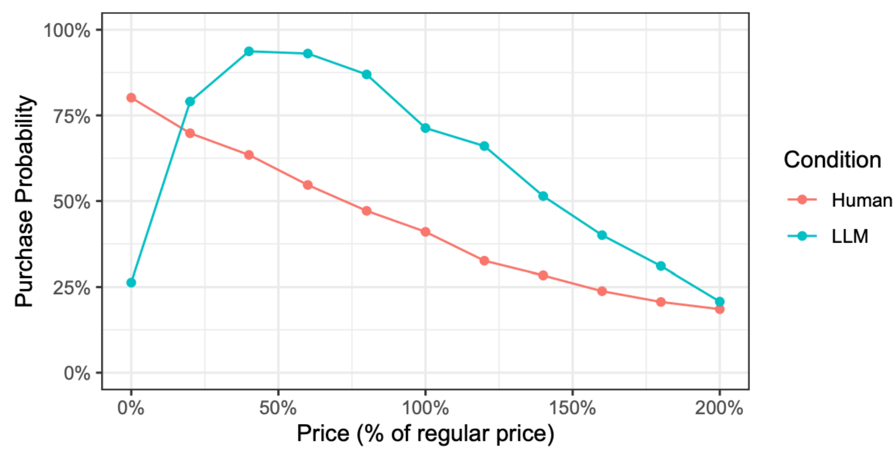


Figure 2: Demand curve elicited from humans vs. LLM, averaged over 40 products

Argyle et al. (2023) introduced an approach for controlling for such covariates. They advocate for running studies on "silicon samples" of digital personas, which in their case were described by demographic variables and political beliefs that were drawn to align with nationally representative samples. This approach has been picked up by the industry and commercialized by several firms (e.g., Synthetic Users, Evidenza). Figure 3 shows that on average, controlling for covariates, such as demographic variables does indeed improve the results in our case compared to not specifying any detail. However, it does not fully resolve the issue. Web Appendix D describes the detailed prompts and covariates used, which involve creating of 500 personas for each {product,category} pair.

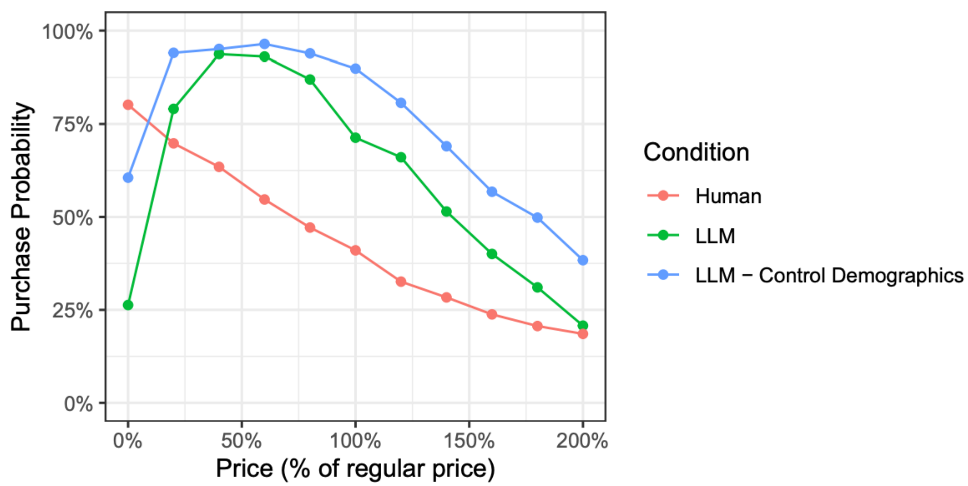


Figure 3: Demand curves elicited by LLM after controlling for demographic variables

Synthetic Personas Beliefs More Coherent than Humans

Synthetic personas distort the structure of human belief systems

Christopher Barrie[1], Roberto Cerina[2]

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[2]Institute for Logic, Language, and Computation, University of Amsterdam

February 25, 2026

Abstract

Large language models (LLMs) are increasingly used as synthetic survey respondents, yet it is unclear whether their belief-system structure matches that of real publics. We compare 28 LLMs to the 2024 General Social Survey (GSS) using 52 attitude items and demographic persona traits. We estimate polychoric correlation matrices and propagate uncertainty in the GSS via bootstrap resampling with multiple imputation. Constraint is measured by the variance share explained by the first principal component and by effective dependence, a determinant-based measure of global linear dependence. Across models, LLM personas exhibit substantially higher constraint than humans; conditioning on persona traits reduces constraint far more for LLMs, indicating greater demographic mediation. Projection onto a shared GSS basis further shows overemphasis of the leading dimension and missing secondary structure. These results caution against treating LLM personas as a reliable foundation for synthetic survey data generation.

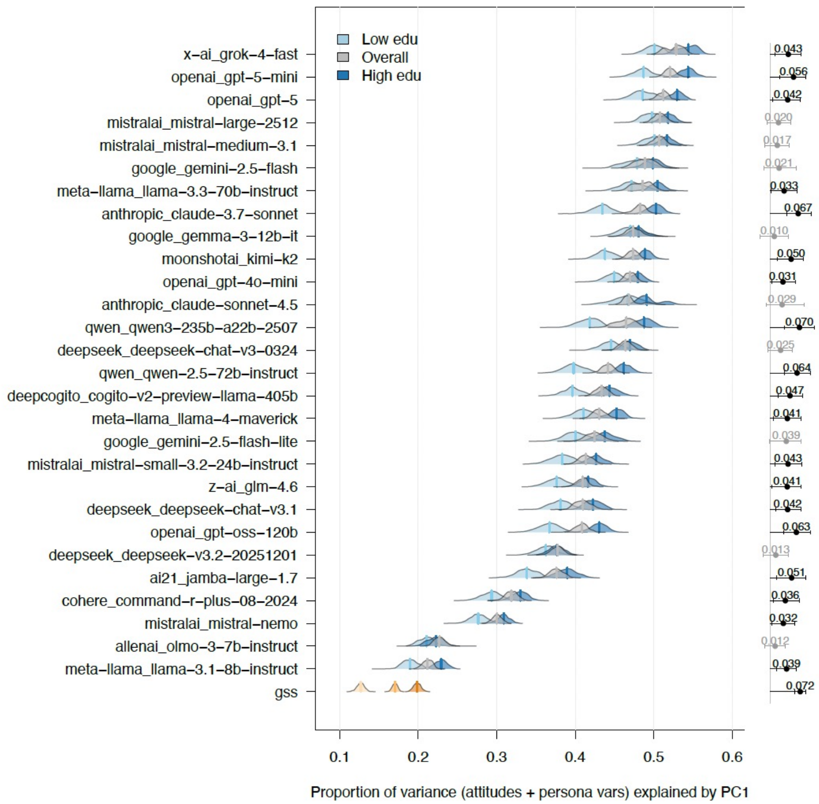

Figure 2: **Dominant-axis constraint by education.** Bootstrap distributions of $\text{PVE}_1 = \lambda_1/p$, the proportion of total system variance explained by the first principal component of the full polychoric correlation matrix (attitudes + persona variables), shown for the GSS and each LLM. Distributions are shown overall and stratified by education (low vs. high). Right-side lollipops show the education gradient $\Delta = \text{PVE}_1(\text{high edu}) - \text{PVE}_1(\text{low edu})$: the point is the median and whiskers denote the 95% bootstrap interval. Lollipops are dark when the 95% interval excludes zero (light otherwise). Larger values indicate more unidimensional belief systems.

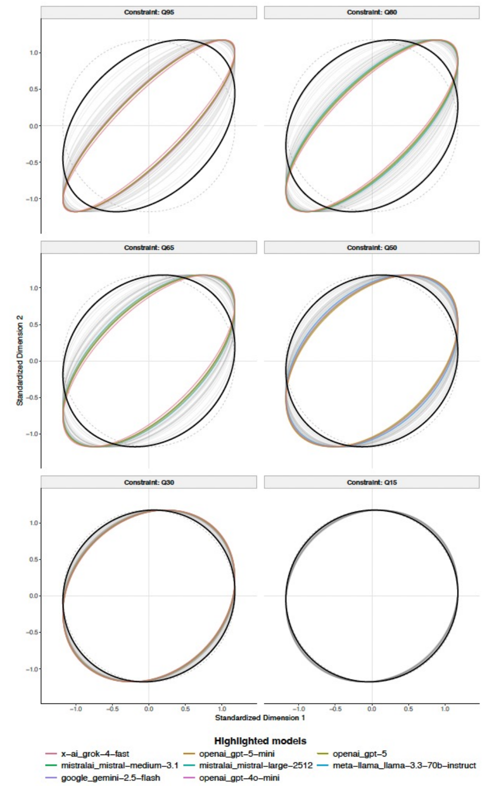

Figure 1: **Faceted Saturn plot summary of system constraint.** Each facet corresponds to a quantile of the absolute correlation distribution $\{|\rho_{j\ell}|\}$ (here Q_{95}, Q_{80}, Q_{65}, Q_{50}, Q_{30}, Q_{15}). For each model and quantile q, we compute the qth quantile of $\{|\rho_{j\ell}|\}$ within each bootstrap draw and take the median across bootstraps to obtain $\rho_{m,q}$. We then plot the 50% probability-mass contour of a bivariate normal distribution with correlation set to $\rho_{m,q}$. The dashed circle is the $\rho = 0$ reference contour at the same probability mass. Faint grey contours show all non-highlighted LLMs; the GSS is shown in bold black. Colored contours highlight model–quantile pairs with large positive $\Delta_{m,q} = \rho_{m,q} - \rho_{\text{GSS},q}$ (threshold $\Delta_{m,q} \geq 0.10$, restricted to the largest deviations per quantile), indicating substantially greater constraint than humans at the same point in the correlation distribution.

Establishing Reliability

LLMs Reproduce Human Purchase Intent via Semantic Similarity Elicitation of Likert Ratings

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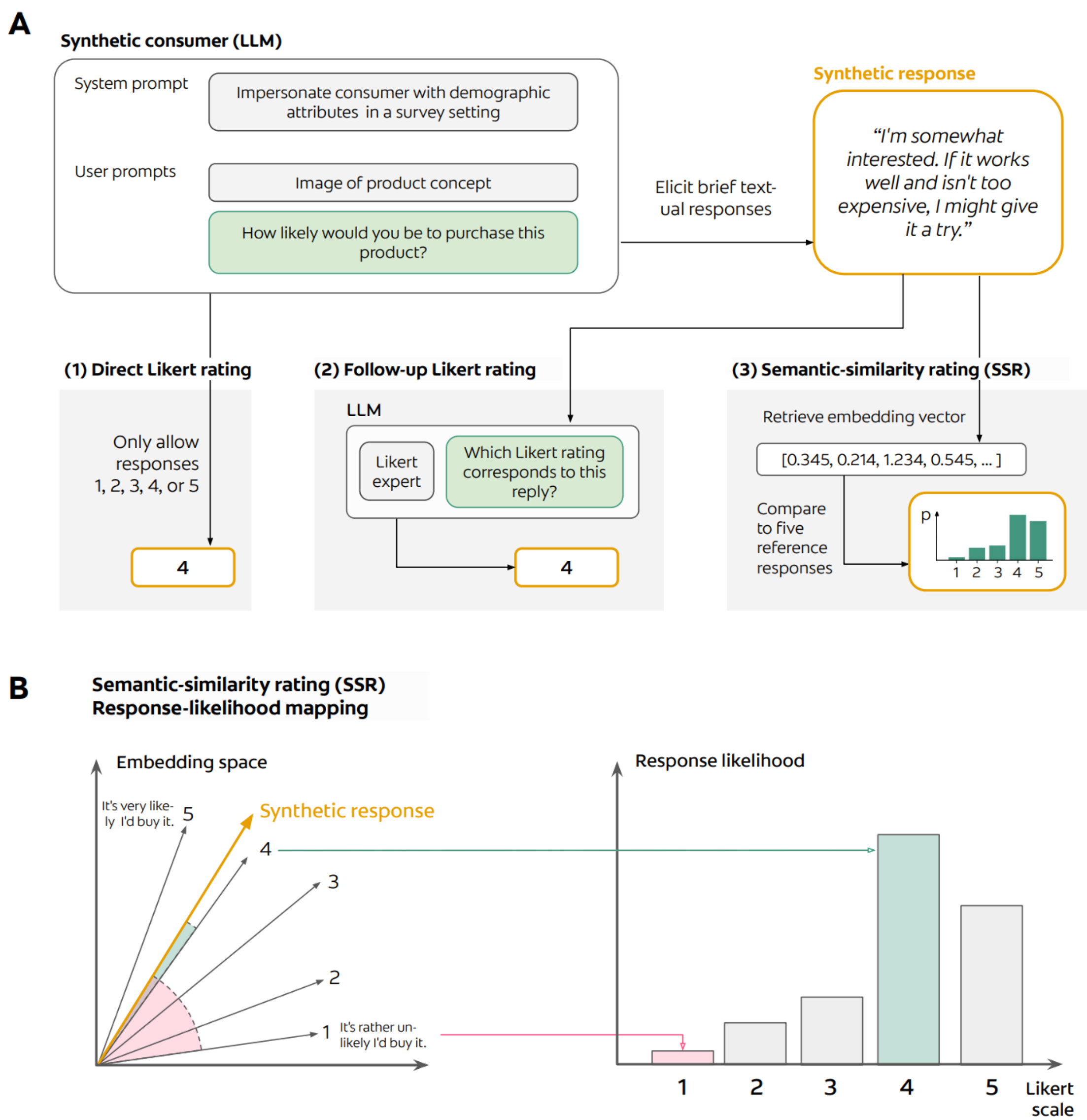
Abstract

Consumer research costs companies billions annually yet suffers from panel biases and limited scale. Large language models (LLMs) offer an alternative by simulating synthetic consumers, but produce unrealistic response distributions when asked directly for numerical ratings. We present semantic similarity rating (SSR), a method that elicits textual responses from LLMs and maps these to Likert distributions using embedding similarity to reference statements. Testing on an extensive dataset comprising 57 personal care product surveys conducted by a leading corporation in that market (9,300 human responses), SSR achieves 90% of human test–retest reliability while maintaining realistic response distributions (KS similarity > 0.85). Additionally, these synthetic respondents provide rich qualitative feedback explaining their ratings. This framework enables scalable consumer research simulations while preserving traditional survey metrics and interpretability.


A
Synthetic consumer (LLM)
System prompt
Impersonate consumer with demographic attributes in a survey setting
User prompts
Image of product concept
How likely would you be to purchase this product?
Elicit brief textual responses
Synthetic response
"I'm somewhat interested. If it works well and isn't too expensive, I might give it a try."
(1) Direct Likert rating
Only allow responses 1, 2, 3, 4, or 5
4
(2) Follow-up Likert rating
LLM
Likert expert
Which Likert rating corresponds to this reply?
4
(3) Semantic-similarity rating (SSR)
Retrieve embedding vector
[0.345, 0.214, 1.234, 0.545, ...]
Compare to five reference responses
p
1 2 3 4 5
B
Semantic-similarity rating (SSR)
Response-likelihood mapping
Embedding space
It's very likely I'd buy it.
5
Synthetic response
4
3
2
1
It's rather unlikely I'd buy it.
Response likelihood
1 2 3 4 5
Likert scale

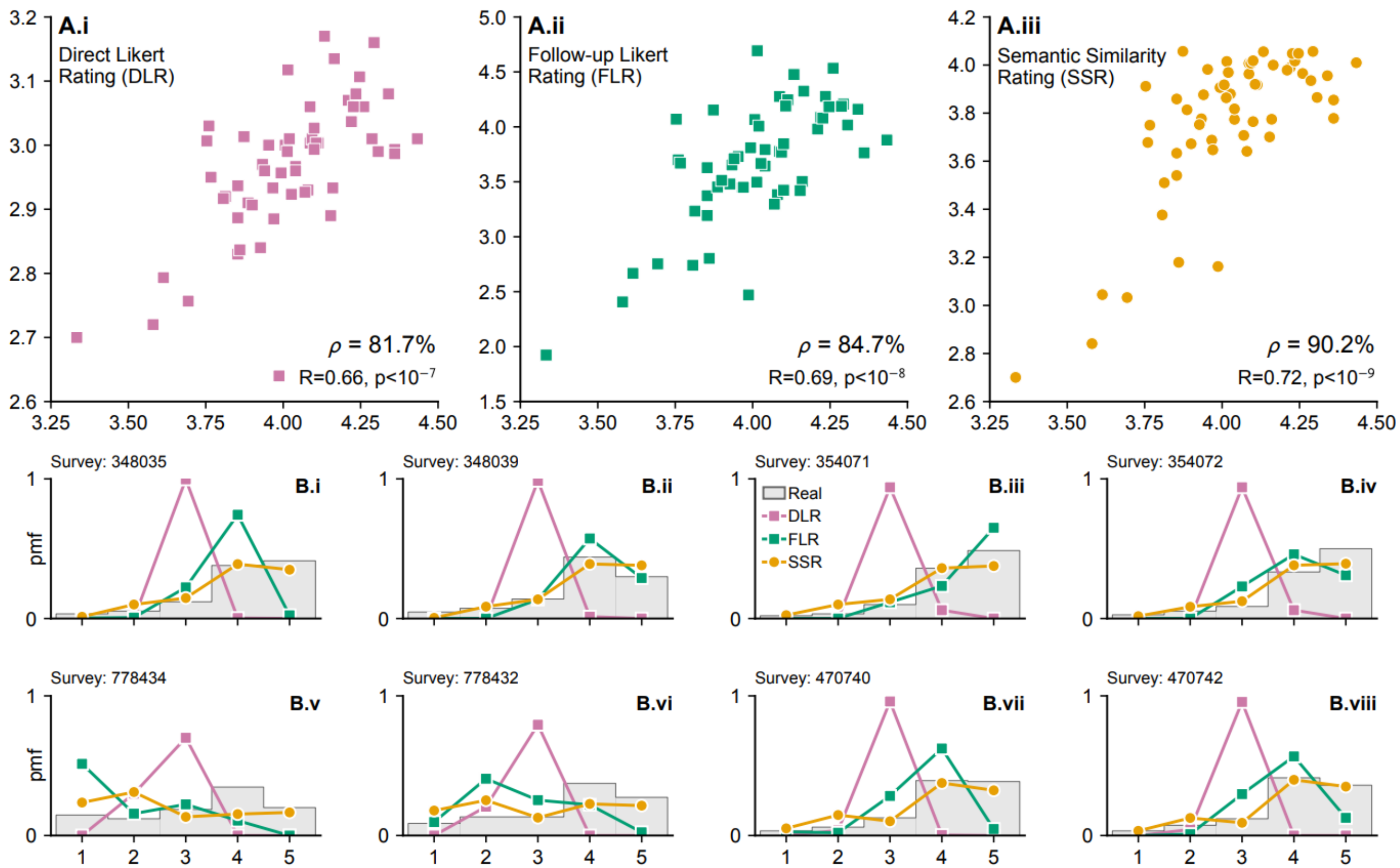


Figure 2: Comparison of real and synthetic surveys based on GPT-4o with $T_{\text{LLM}} = 0.5$. (A) Mean purchase intent comparison for (A.i) Direct likert ratings (DLRs), (A.ii) textual elicitation with follow-up Likert ratings (FLRs) and (A.iii) semantic similarity ratings (SSRs). (B) Eight example survey response distributions for real surveys and the corresponding synthetic surveys based on DLR, FLR, and SSR, respectively.

Simulations: do they work?

- The promise: LLMs can stand in for economic agents in experiments
  - Allows rapid iteration on consumer choice before running expensive human trials
  - Synthetic consumer panels can revolutionize market research
- Challenges:
  - Choice homogeneity: agent demand may collapse on a few set of “modal products”
  - Focal bias: LLMs overweight covariates in their attention
  - Demographic gaps: accuracy drops for partisan groups and polarized topics
  - Individual accuracy hard; even group accuracy challenging

How to make them work?

- How to get them to work better:
  - Applied to textual statements by people
  - Converting synthetic responses to Likert scales
  - Fine-tuning
  - Demographic matching
- Other considerations:
  - **Interpolation** works better than **extrapolation**
  - Cases when the LLM can “pattern match” to something in its training dataset or sample, rather than outlier events or drastic shifts
  - Great for cases with a low cost of error and trialing purposes

Where is the moat?

- Proprietary data, not proprietary analysis
  - If analytic engine is available to everyone, scarce input becomes the data
  - Private markets look more appealing relative to public markets, though are likely facing a "Bloomberg" type leveling information shock too
- Speed of action, not speed of reading
  - If you speed up the analyst reading but not the investment committee process or portfolio management decision, that doesn't help
- Judgment under uncertainty
  - So far, humans retain an edge when soft information, broader context, new situations break, or private information matter